USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

1) Please fill out the Engineering Change Request Form on the following pages completely:
   a) Detail the names and contact details for each of the ECR contributors
   b) Update the ECR Title
   c) Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
2) For each section/table/figure to be updated:
   a) Detail the section number, starting page and figure/table number to be updated as appropriate.
   b) Detail existing text under "From Text"
   c) Detail changed text under "To Text"
3) Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step 1)b)
4) Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
   a) This ECR will then be reviewed by the Power Delivery Compliance Working Group.
   b) Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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City: _____ State/Province: _____

Country: _____ Zip/Postal Code: _____

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: VPD test update – TEST.PD.PROT.ALL.4 Test
Applied to: USB PD CTS Q3, 2026

Brief description of the functional changes proposed:
Add wait time “teMarkerReady” after a tester issued cable reset.

Benefits as a result of the proposed changes:
eMarker DUTs tested per spec definition after a Cable Reset.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
Tester to update the corresponding tests per the ECR updates.

An analysis of the Vendor Info File (VIF) implications:
None

Actual Change Requested

TEST.PD.PROT.ALL.4

From Text:

5. Immediately after sending *GoodCRC* (to the *Discover Identity* ACK, *Sink_Capabilities* or *Reject*), the Tester sends a Hard-Reset signal after *tInterframeGap* (note that the second run through is a *Cable Reset* Signal for a Cable Plug UUT).
6. The test proceeds depending on the UUT type/role:
    a. If the UUT is a Sink, the Tester drives V_{BUS} to *vSafe0V* and then restores V_{BUS} to *vSafe5V*. The Tester transmits *Source_Capabilities* Messages with single PDO (5V @ 500mA, Source only) immediately after V_{BUS} reaches *vSafe0V* min. The Tester checks that the UUT sends a *Request* Message. [TEST.PD.PROT.ALL.4#4] Once it receives a *Request* Message, it proceeds using common procedure COMMON.PROC.PD.10. The Tester then waits **Error! Reference source not found.** to respond to messages from the UUT.
    b. If the UUT is a Source:
        i. The Tester checks that the UUT drives V_{BUS} to *vSafe0V* and then restores V_{BUS} to *vSafe5V*. [TEST.PD.PROT.ALL.4#5]
        ii. The check fails if *Source_Capabilities* is not sent by the UUT within *tFirstSourceCap*. [TEST.PD.PROT.ALL.4#6] Once the Tester receives *Source_Capabilities* Message, the Tester establishes a PD contract using common procedure COMMON.PROC.PD.11. The Tester then waits **Error! Reference source not found.** to respond to messages from the UUT.
    c. If the UUT is a Cable Plug, the Tester checks the response depending on the **run number**. [TEST.PD.PROT.ALL.4#7]
        i. If the Tester has sent a *Hard Reset* (**first run**), the Tester drives VBUS to *vSafe0V* and then restores VBUS to *vSafe5V*. The Tester run VCONN power cycles process. The Tester transmits *Discover Identity* REQ every *tDiscoverIdentity* until either it receives a *GoodCRC* followed by a *Discover Identity* ACK or it has transmitted *Discover Identity* REQ 3 times. The check fails when the Tester has transmitted *Discover Identity* REQ for 3 times.
        ii. If the Tester has sent a *Cable Reset* (**second run**), the Tester transmits *Discover Identity* REQ every *tDiscoverIdentity*. The check fails if the UUT does not sends a *GoodCRC* followed by a *Discover Identity* ACK.

To Text:

5. Immediately after sending *GoodCRC* (to the *Discover Identity* ACK, *Sink_Capabilities* or *Reject*), the Tester sends a Hard-Reset signal after *tInterframeGap* (note that the second run through is a *Cable Reset* Signal for a Cable Plug UUT).
6. The test proceeds depending on the UUT type/role:
    a. If the UUT is a Sink, the Tester drives V_{BUS} to *vSafe0V* and then restores V_{BUS} to *vSafe5V*. The Tester transmits *Source_Capabilities* Messages with single PDO (5V @ 500mA, Source only) immediately after V_{BUS} reaches *vSafe0V* min. The Tester checks that the UUT sends a *Request* Message. [TEST.PD.PROT.ALL.4#4] Once it receives a *Request* Message, it proceeds using common procedure COMMON.PROC.PD.10. The Tester then waits **Error! Reference source not found.** to respond to messages from the UUT.
    b. If the UUT is a Source:
        i. The Tester checks that the UUT drives V_{BUS} to *vSafe0V* and then restores V_{BUS} to *vSafe5V*. [TEST.PD.PROT.ALL.4#5]

ii. The check fails if *Source_Capabilities* is not sent by the UUT within *tFirstSourceCap*. [TEST.PD.PROT.ALL.4#6] Once the Tester receives *Source_Capabilities* Message, the Tester establishes a PD contract using common procedure COMMON.PROC.PD.11. The Tester then waits **Error! Reference source not found.** to respond to messages from the UUT.

c. If the UUT is a Cable Plug, the Tester checks the response depending on the **run number**. [TEST.PD.PROT.ALL.4#7]

i. If the Tester has sent a *Hard Reset* (**first run**), the Tester drives VBUS to *vSafe0V* and then restores VBUS to *vSafe5V*. The Tester run VCONN power cycles process. The Tester transmits *Discover Identity* REQ every *tDiscoverIdentity* until either it receives a *GoodCRC* followed by a *Discover Identity* ACK or it has transmitted *Discover Identity* REQ 3 times. The check fails when the Tester has transmitted *Discover Identity* REQ for 3 times.

ii. If the Tester has sent a *Cable Reset* (**second run**), the Tester shall wait at least *teMarkerReady<Table 4-29 eMarker Timing - Type-C specification>* after sending the *Cable Reset*, then transmits *Discover Identity* REQ every *tDiscoverIdentity*. The check fails if the UUT does not send a *GoodCRC* followed by a *Discover Identity* ACK.

Spec References:

USB PD R3.2 v1.2

7.1.4. Cable Reset

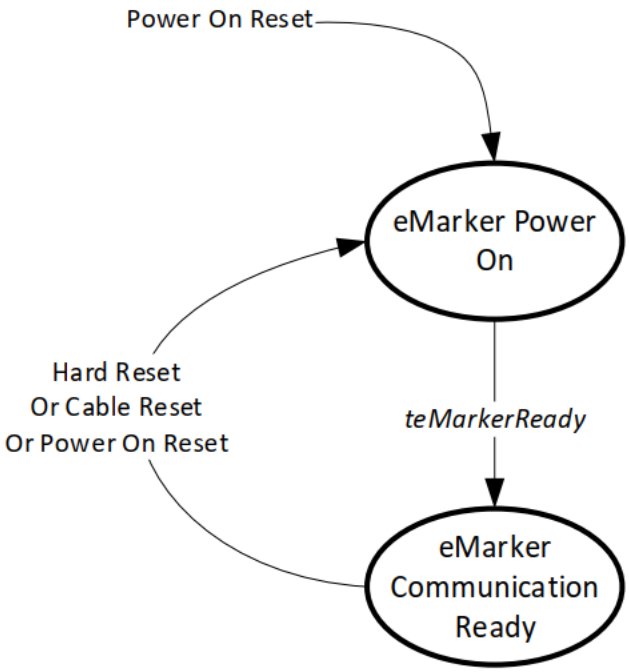
Cable Resets are signaled by an ordered set as defined in Table 5.4. Both the sender and recipient of Cable Reset Signaling **Shall** reset their respective Protocol Layers. The Cable Plugs **Shall** perform the equivalent of a power cycle returning to their initial power up State. This allows the Port Partners to be in a State where they can re-establish USB PD communication.

Only a DFP **Shall** generate Cable Reset Signaling. The DFP must be supplying VCONN prior to a Cable Reset. If VCONN has been turned off the DFP **Shall** turn on VCONN prior to generating Cable Reset Signaling. If there has been a VCONN Swap and the UFP is currently supplying VCONN, the DFP **Shall** perform a VCONN Swap such that it is supplying VCONN prior to generating Cable Reset Signaling. A DFP **Shall** only generate Cable Reset Signaling within an Explicit Contract.

A Cable Reset **Shall** cause all Active Modes in the Cable Plugs to be exited (see Section 8.6.4").

USB Type-C Spec R2.5

Figure 4-20 Cable eMarker State Diagram



4.9.1 Parameter Values

Table 4-24 provides the power on timing requirements for an eMarker to be ready to communicate.

Table 4-24 eMarker Timing

	Maximum	Description
teMarkerReady	50 ms	The time between the application of VCONN or VBUS, whichever is later, until an eMarker ***shall*** be ready for communication.